

Project-Definition Pack — Index

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

Project Identity: A Kaggle-friendly, interpretable ML research pipeline for identifying candidate regulatory switch-like patterns in non-coding DNA, built on public datasets such as ENCODE.

This directory contains the complete project-definition pack. Each document is self-contained but cross-referential. Together they form the operating manual for the project.

Document Index

#	Document	Purpose	File
01	Project Charter	Problem, vision, mission, goals, constraints	01-project-charter.md
02	Product Requirements Document	Features, use cases, requirements, acceptance criteria	02-product-requirements-document.md
03	Technical Design Document	Architecture, pipelines, stack, execution patterns	03-technical-design-document.md
04	Research Design Document	Hypotheses, experiments, baselines, evaluation	04-research-design-document.md
05	Dataset Strategy	Data sources, labels, splits, preprocessing, ethics	05-dataset-strategy.md
06	Modeling Roadmap	Phased model plan, decision gates	06-modeling-roadmap.md
07	Compute & Feasibility Memo	Why feasible, what Kaggle supports, bottlenecks	07-compute-feasibility-memo.md
08	Experiment Tracking & MLOps Lite	Conventions, reproducibility, versioning	08-experiment-tracking-mlops.md
09	Risk Register	Risks, mitigations, fallback plans	09-risk-register.md
10	Roadmap & Milestones	1/3/6/12-month plans, success criteria	10-roadmap-milestones.md
11	Glossary & Project Memory	Definitions, locked decisions, forbidden scope	11-glossary-project-memory.md
12	Contributor Onboarding Brief	How to join, what's decided, expectations	12-contributor-onboarding.md

How to Use This Pack

1. **New contributors** — start with Document 12 (Onboarding Brief), then read Document 01 (Charter) and Document 11 (Glossary).
2. **Technical contributors** — additionally read Documents 03, 05, 06, and 08.
3. **Research contributors** — additionally read Documents 04, 05, and 06.
4. **Project leads / advisors** — read Documents 01, 02, 09, and 10 for governance and milestones.

Scope Lock (Reproduced for Visibility)

- **Domain:** Non-coding regulatory genomics.
 - **Core Objective:** Discover and interpret candidate regulatory switch-like patterns in non-coding DNA.
 - **Early Execution Environment:** Kaggle-friendly.
 - **Early Model Philosophy:** Interpretable and lightweight first.
 - **Large Genomic Transformer Training:** Out of scope for initial build because of compute limits.
 - **Preferred Data Sources:** Public datasets such as ENCODE.
 - **Long-Term Evolution:** Stronger models, richer interpretation, broader generalization — but only after a working baseline exists.
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